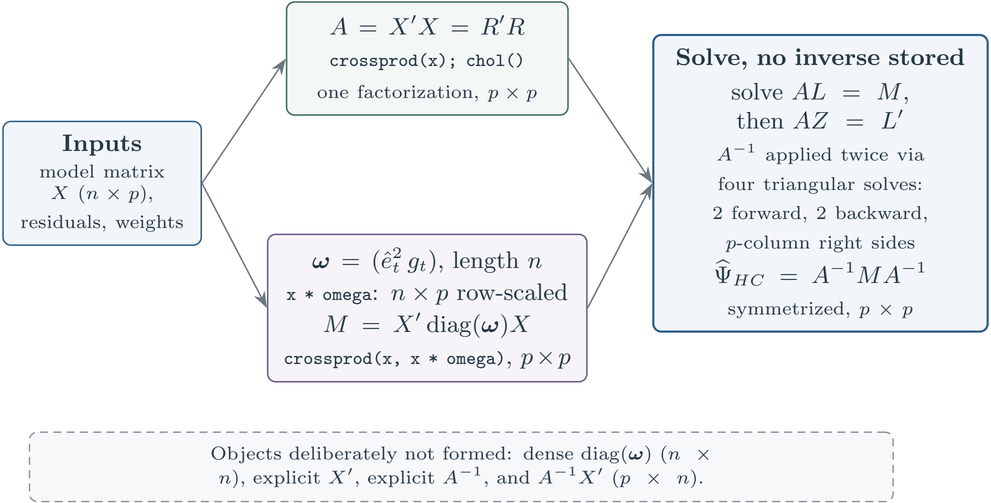
Inputs
model matrix
X $(n \times p)$,
residuals, weights
$A = X'X = R'R$
`crossprod(x); chol()`
one factorization, $p \times p$
$\boldsymbol{\omega} = (\hat{e}_t^2\, g_t)$, length n
`x * omega`: $n \times p$ row-scaled
$M = X' \operatorname{diag}(\boldsymbol{\omega}) X$
`crossprod(x, x * omega)`, $p \times p$
Solve, no inverse stored
solve $AL = M$,
then $AZ = L'$
A^{-1} applied twice via
four triangular solves:
2 forward, 2 backward,
p-column right sides
$\widehat{\Psi}_{HC} = A^{-1} M A^{-1}$
symmetrized, $p \times p$
Objects deliberately not formed: dense $\operatorname{diag}(\boldsymbol{\omega})$ $(n \times n)$, explicit X', explicit A^{-1}, and $A^{-1}X'$ $(p \times n)$.